SOE103 C Programming and Software Engineering – Group Project

Assessment Number	3 (group project)
Contribution to Overall Marks	15%
Issue Date	November 2, 2026
Submission Deadline	December 13, 2026, 23:59

Assessment Objective

This assessment aims to evaluate students' ability to develop a significant software solution to a real-world problem by working as a member of a team. Your team will be given a vague specification and is expected to deliver a software product using the C programming language, which meets the specifications before the due date. This size and type of project is suitable for development in a modular format, and so teams are encouraged to devise program structures that allow various parts of the code to be developed independently by each team member. Being a team player means you are expected not only to apply the knowledge gained during the lectures, laboratory classes, and assignments to specify, design, implement, test, and document your own code, but also to cooperate with your teammates so that the whole project will be delivered on time with good quality.

Grouping

There are 447 students enrolled in this module, and you will be divided into groups strictly consisting of **5 students**. Groups will be formed in two stages as follows: Firstly, **students are strongly recommended to choose their own group members**. Students failing to form a group will then be randomly assigned to a group. Each randomly formed group will strictly have 5 students unless there are no more students available. Each group will then develop a Pizza Ordering System following the attached specification using the C programming language.

Final Deliverables

Each group should submit the following by the group leader (only 1 submission per group):

1. A report (PDF file) based on the Software Development Process applied throughout this semester:
   a) Problem Specification (formulate the problem generally) (10%).
   b) Analysis: interpret the vague software requirements provided by your design brief and determine a very clear specification for your software design. (10%)
   c) Algorithm Design: explain how your program is structured in terms of functional blocks and describe what each block does. A flow diagram may be useful here, showing how functions interconnect and what data is passed between them. (10%)
   d) Testing: explain how the program has been tested and verified. Ensure any usernames or passwords required to test the program are listed in a table. (10%)
2. All C source code (.c and .h files) and the final executable demonstration file (.exe). The source code must be appropriately commented. Include any data files needed for execution. (50%)

3. Contribution form (The contribution form is at the end of the file). The group should agree on the percentage of contribution of each member to the project and submit one copy of the form signed by all members. Different individual contributions to the group project result in different individual marks.

NOTE: Different individual contributions to the project may lead to different marks for different members of the same group. The module leader may call group members for a short oral presentation if necessary.

Submission Procedure

All the above-mentioned files/deliverables (report, source code (.c and .h) files, data files, executable file(.exe), manual and contribution form) must be compressed into **a single file (.rar or .zip)**. **The group leader** (identified on the group contribution form) must submit this single compressed file on Learning Mall using his/her account.

NOTE: Each group should only submit **ONE** copy on Learning Mall by the group leader. Make sure your report has a title page and ensure **ALL** group members' names and ID numbers are on it.

Marking Scheme

The general marking scheme is shown as follows:

Documentation	**(50%)**
Overall Report Quality	10%
Problem Specification	10%
Analysis	10%
Algorithm Design	10%
Testing	10%
Coding	**(50%)**
Implementation/coding style	40%
Robustness	10%

Important notes:

The university's late submission policy will be applied.

The University Academic Integrity policy will be applied rigorously.

You can find information relating to both of these in eBridge.

General Guidelines

The project descriptions are deliberately given in the form of simple customer specifications, which (as in the real world) are incomplete and often ambiguous, rather than a set of exact functional specifications. The group members should work methodically together (as the developers in a real-world software project would) to:

1. Analyse and formalise the customer specifications (at this stage, the various design choices and the software features can be subject to the group's creativity).
2. Design and decompose the functional and programmatic aspects of the problem and allocate constituent tasks to each group member. You are expected to use a top-down design, which can then be modularised so that the tasks for each member can be clearly determined.
3. Implement the product with frequent meetings to report progress and decisions to each other and re-evaluate the agreed courses of action.
4. Implement test procedures, debug and correct the program. Each program module should be independently testable. Testing of each module and the program as a whole should be performed.
5. Finalise the deliverables.

The specifications are only basic, and most of the design choices should be made during your group meetings. The disambiguation of the customer specifications can be based on the student's logic and real-life experience.

Assessment will be based on whether the product/program offers reasonable functionality and features (for the group size, allocated time, and project difficulty), its design quality, flexibility, robustness, software bugs, and other stated deliverables.

If the group cannot implement all of the system features mentioned, it is better to have a few features fully working without run-time crashes than none of the required features working properly due to bugs or disrupting ripple effects between modules in the project. However, the corresponding marks deduction will be applied depending on the missing features.

If any group issues arise during the project, it is important to raise these with the module leader as soon as they do so that resolutions can be found quickly. It is important to work with your group and to share the work accordingly. All group members should be responsible for some of the coding. Individuals may not produce your own work for submission independent of the group.

Project: Pizza Ordering System

Overall description:

Your team is employed by a new pizza shop to develop the software for an in-store terminal where customers can order their pizza.

Customer specifications:

The implemented ordering system should be able to provide facilities to:

• Allow the various types of pizza on the menu (e.g. Thin and Crispy Crust, and Hand Tossed) to be ordered, where each pizza can be customised by different size (e.g. 10", 12", 14", and 16") and topping (e.g. Supreme, Hawaiian, Margherita, and BBQ Chicken).

• Determine the cost of the pizza ordered based on the type, size, and topping.

• Register and edit customer accounts, identifiable by a unique account number, which can be pre-charged with money. Customer information should include name, telephone, order history, and available balance.

System Users

The system should be able to provide the functionality for the different users listed below:

- **Shop manager who will be able to:**
  - o Edit the pizza menus.
  - o Set/alter charges for the pizza.
  - o Add/remove/edit customer information.
  - o Provide statistics related to pizza sales.
- **Customer Self Service will be able to:**
  - o Login to the account and place an order.
  - o Review order history.
  - o Deposit money on the account.

SOE103 Group Project
Contribution Form

Group Leader Name: **Student ID:**

Name	Student ID	Contribution (%)	Signature
1.			
2.			
3.			
4.			
5.			
	Total	**100%**	